INCH-POUND

MIL-M-28787/208
AMENDMENT 4
31 March 1993
SUPERSEDING
AMENDMENT 3
19 February 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAMMABLE TIMER, DIGITAL, KEY CODE HQE

This amendment forms a part of Military Specification MIL-M-28787/208(EC), dated 12 June 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

In bottom right hand corner, delete "FSC MISC" and substitute "FSC 5963".

PAGE 3

1.2, Module description, Propagation delay time, Timers 1 and 3, tPLH (C_L = 50 pF), Z to T: Delete "1,500 ns" and substitute "2,500 ns".

1.2, Module description, Propagation delay time, Timer 2, tPLH (C_L = 50 pF), P to T: Delete "2,100 ns" and substitute "2,500 ns".

1.2, Module description, Propagation delay time, Timer 4, tPLH (C_L = 50 pF), P to T: Delete "1,900 ns" and substitute "2,500 ns".

1.3, Module function: Delete first sentence in its entirety.

PAGE 9

3.5.8, second line: Delete "M28787/208" and substitute "M28787/208-1".

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PAGES 10 AND 11

4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 11

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A and B are stress tests and shall be performed only at 25 ± 5°C. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A and B where applicable, followed by the remaining tests in the sequence."

PAGE 13

TABLE IV, Test condition P1, Min columns: Delete "5" (four places) and substitute "2" (four places).

TABLE IV, Test condition P3, Min columns: Delete "5" (four places) and substitute "0.5" (four places).

TABLE IV, Test condition P4, Min columns: Delete "7" (four places) and substitute "3" (four places).

PAGE 15

TABLE IV, Test conditions Y3 and Y4, Max columns: Delete "1,500" (four places) and substitute "2,500" (four places).

TABLE IV, Test condition Y5, Max columns: Delete "2,100" (four places) and substitute "2,500" (four places).

TABLE IV, Test condition Y6, Max columns: Delete "1,900" (four places) and substitute "2,500" (four places).

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PAGE 17

Add the following new paragraph:

"6.3 Cross reference of classification. Part number M28787/208-1 supersedes part number M28787/208. The two parts are interchangeable.

PAGE 26

FIGURE 8, NOTE: Delete and substitute "NOTE: Counter = HP5302A, or equivalent. Set counter to measure period."

PAGE 28

FIGURE 9: Delete and substitute new figure 9 as printed on pages 6 and 7 of this amendment.

PAGE 31

FIGURE 10: Delete and substitute new figure 10 as printed on pages 8 and 9 of this amendment.

PAGE 33

FIGURE 11: Delete and substitute new figure 11 as printed on pages 10 and 11 of this amendment.

PAGE 36

FIGURE 12: Delete and substitute new figure 12 as printed on pages 12 and 13 of this amendment.

PAGE 48

FIGURE 19: Delete and substitute new figure 19 as printed on page 14 of this amendment.

PAGE 53

TABLE XVIII, Test condition P1, Min column: Delete "5" and substitute "2".

TABLE XVIII, Test condition P3, Min column: Delete "5" and substitute "0.5".

TABLE XVIII, Test condition P4, Min column: Delete "7" and substitute "3".

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PAGE 56

FIGURE 23: Delete and substitute new figure 23 as printed on pages 15 and 16 of this amendment.

PAGE 58

FIGURE 24, NOTE: Delete and substitute "NOTE: Counter = HP5302A, or equivalent. Set counter to measure period."

PAGE 60

FIGURE 25, NOTE 2: Delete and substitute "2. Counter = HP5302A, or equivalent. Set counter to measure period."

PAGE 63

FIGURE 27, NOTE 2: Delete and substitute "2. Counter = HP5302A, or equivalent. Set counter to measure period."

PAGE 70

FIGURE 31: Delete and substitute new figure 31 as printed on pages 17 and 18 of this amendment.

PAGE 78

TABLE XXVII, Test condition Y3, Max column: Delete "1,500" and substitute "2,500".

TABLE XXVII, Test condition Y4, Max column: Delete "1,500" and substitute "2,500".

TABLE XXVII, Test condition Y5, Max column: Delete "2,100" and substitute "2,500".

TABLE XXVII, Test condition Y6, Max column: Delete "1,900" and substitute "2,500".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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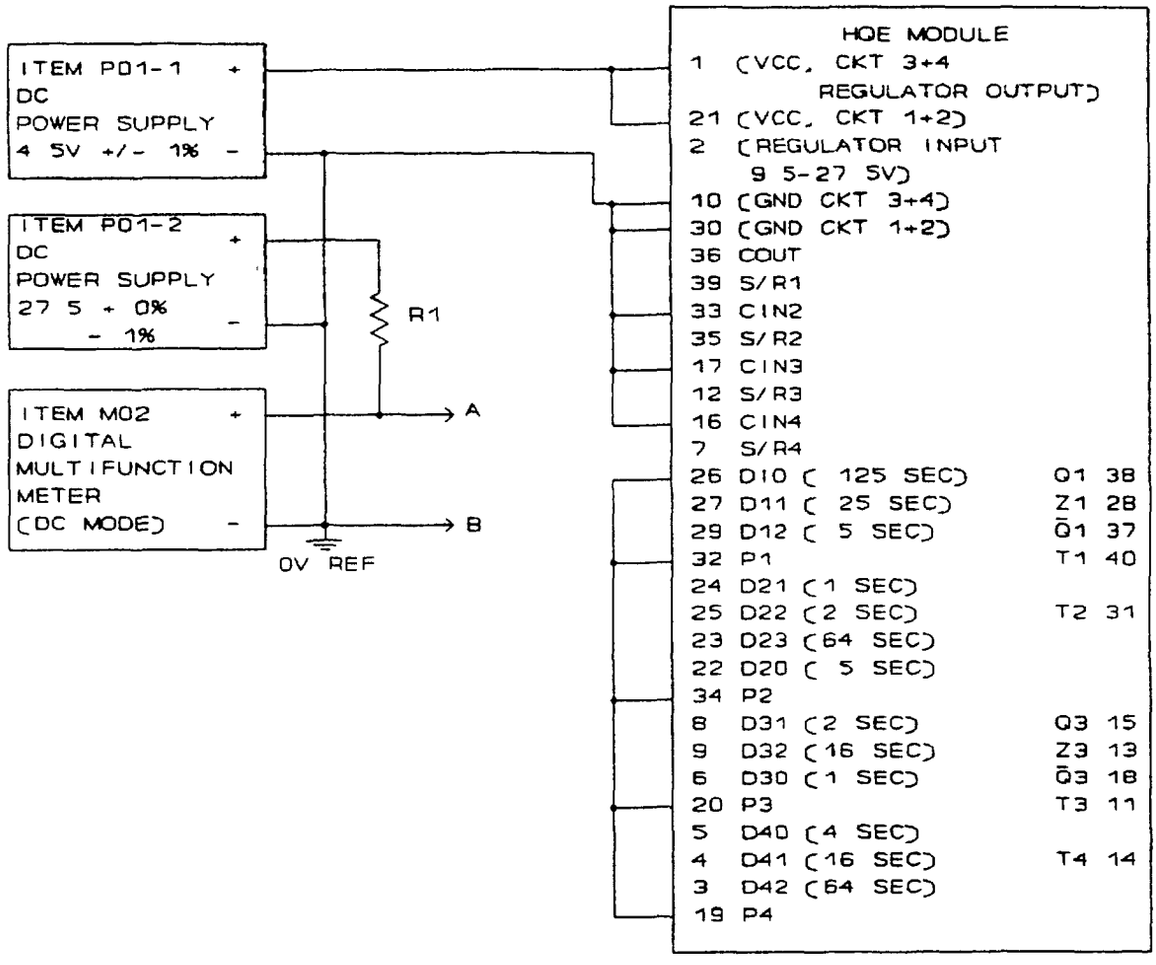
Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0179)

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CAUTION:

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 9. T output voltage low level, VCC = 4.5, test circuit for table VII.

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NOTES: (Continued)

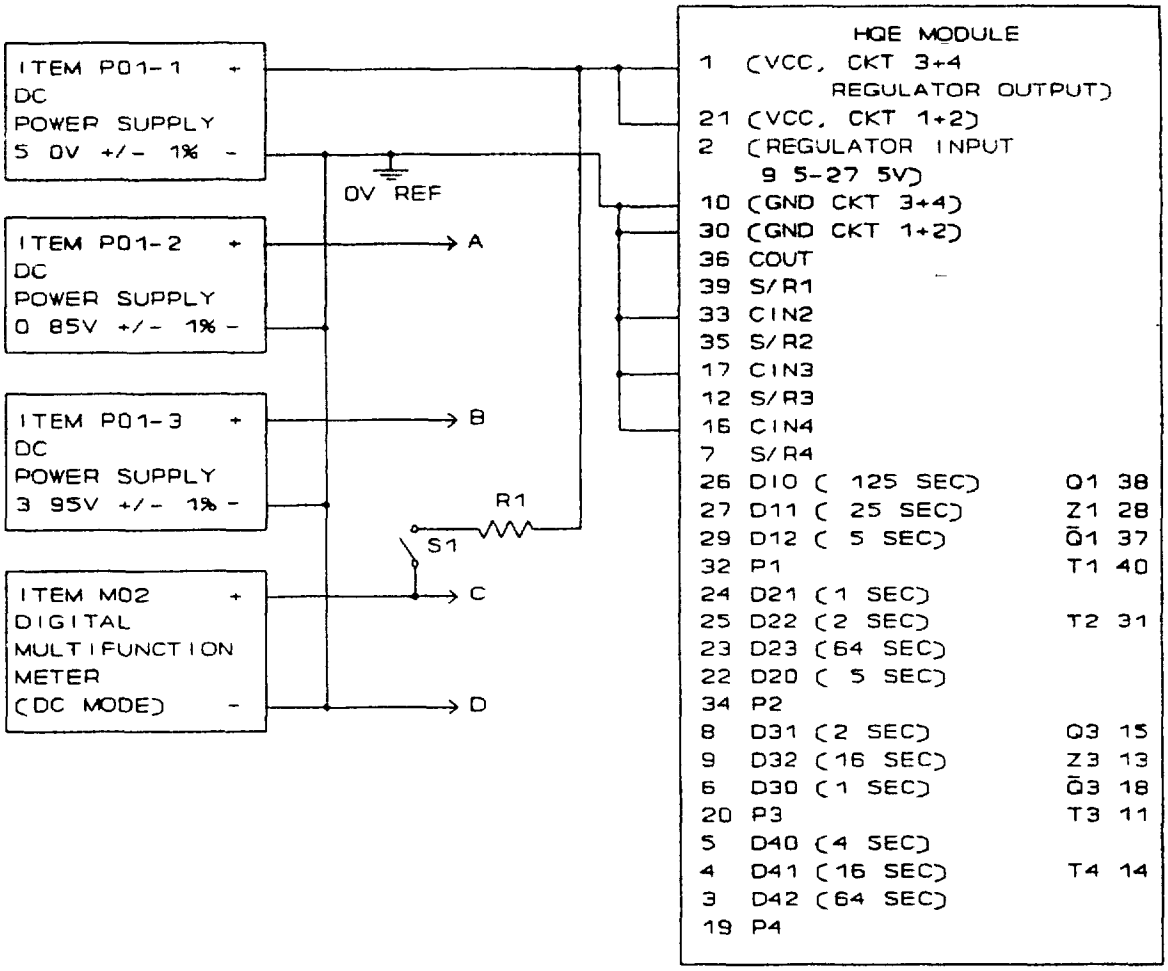
1. R1 = 1.1 kΩ ± 1%, 1 W.

2. Make pin to pin connections prior to connecting points A or B.

3. For test conditions C4-C6, make the output measurement a minimum of 0.135 seconds after pin 39 has been connected to point B.

FIGURE 9. T output voltage low level, VCC = 4.5, test circuit for table VII - Continued.

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CAUTION:

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 10. VOUT, T, D, Q, and $\overline{Q}$ voltage, low level, test circuit for table VIII.

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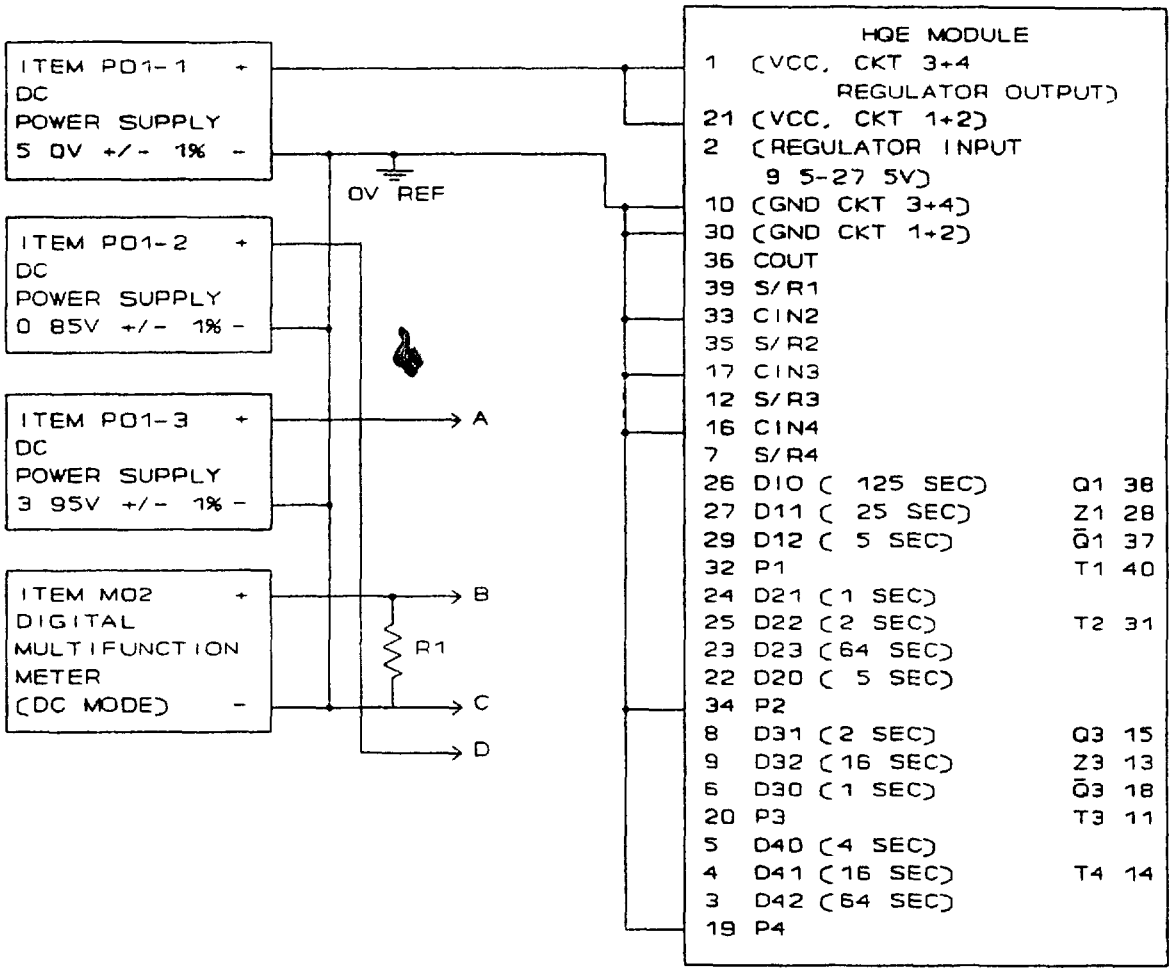
NOTES: (Continued)

1. For test conditions D1-D4, R1 = 187 Ω ± 1%, 1/4 W, S1 closed.
2. For test conditions D5-D21, R1 value ± 1%, 1/8 W listed below, S1 closed.
3. For test conditions D22-D25, S1 open.
4. S1 = SPST.

TEST COND	0° TO 25°C	60° TO 80°C
D5 - D7	18.2 kΩ	26.1 kΩ
D8 - D17	45.3 kΩ	49.9 kΩ
D18 - D21	1.54 kΩ	2.15 kΩ

FIGURE 10. VOUT, T, D, Q, and $\overline{Q}$ voltage, low level, test circuit for table VIII - Continued.

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CAUTION:

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 11. Q, $\overline{Q}$, and D output voltage, high level, test circuit for table IX.

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NOTES: (Continued)

1. For test conditions E1-E7, R1 values ± 1%, 1/8 W listed below.

TEST COND	0° TO 25°C	60° TO 80°C
E1 - E4	9.31 kΩ	12.1 kΩ
E5 - E7	28.0 kΩ	40.2 kΩ

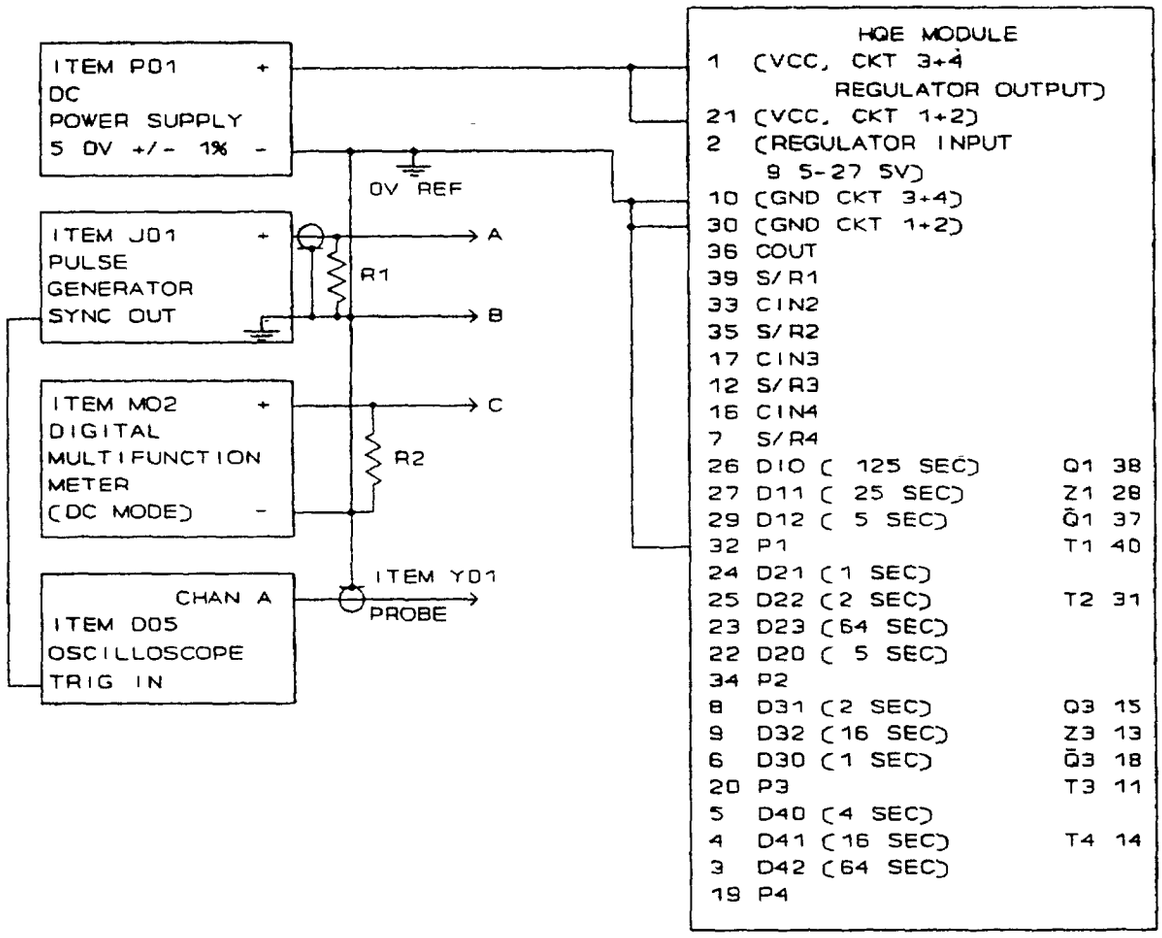
2. Test E5, make measurement .125 second after D is applied to pin 39.

3. Test E6, make measurement .25 second after D is applied to pin 39.

4. Test E7, make measurement .50 second after D is applied to pin 39.

FIGURE 11. Q, $\overline{Q}$, and D output voltage, high level, test circuit for table IX - Continued.

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CAUTION:

Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 12. D output voltage, high level, test circuit for table X.

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NOTES: (Continued)

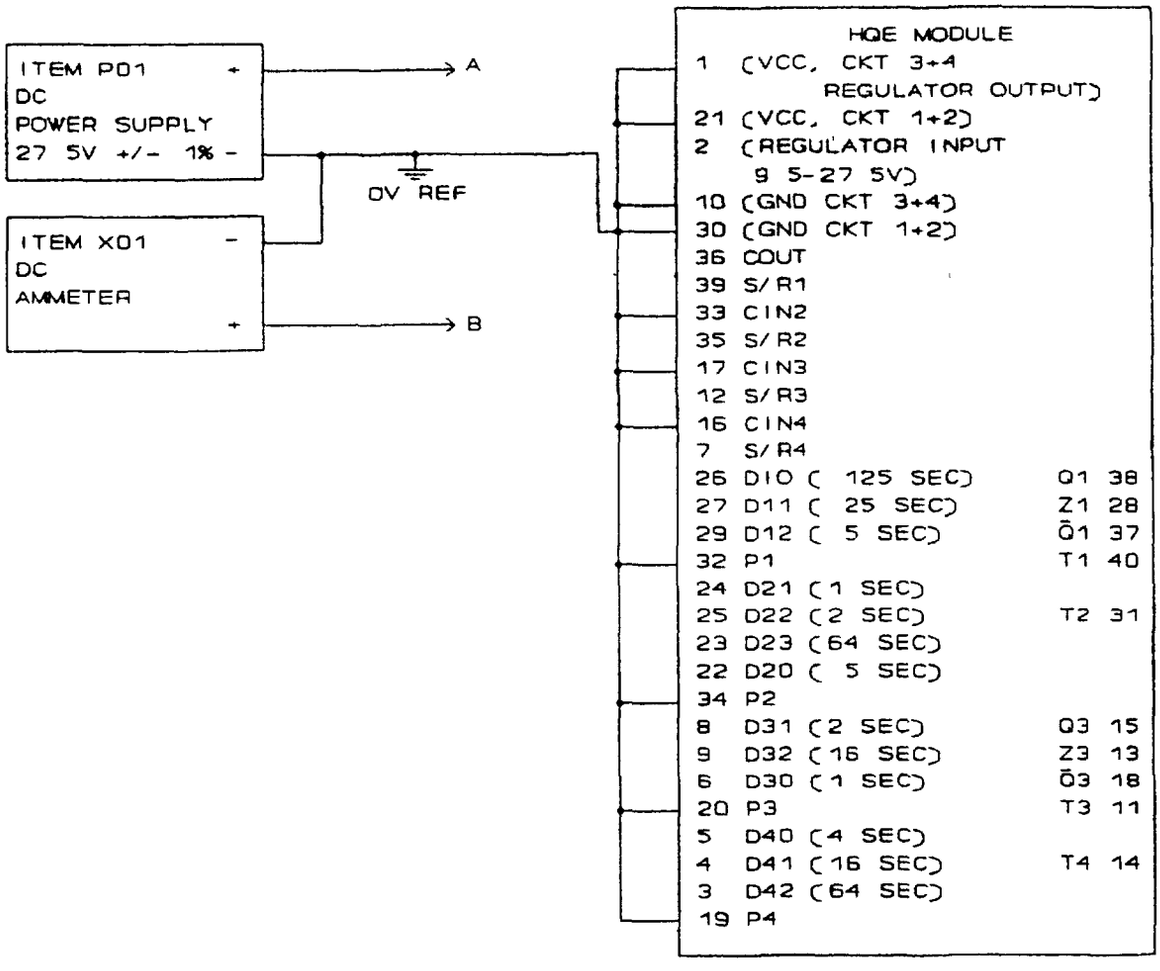
1. R1 = 51 Ω ± 5%, 1/4 W, R2 at 0°-25°C = 46.4 kΩ ± 1%, 1/8 W, R2 at 60°-80°C = 64.9 kΩ ± 1%, 1/8 W.

2. Make all other circuit connections before connecting point B.

3. Make measurement 3 μs after connecting point B.

FIGURE 12. D output voltage, high level, test circuit for table X - Continued.

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CAUTION:
Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

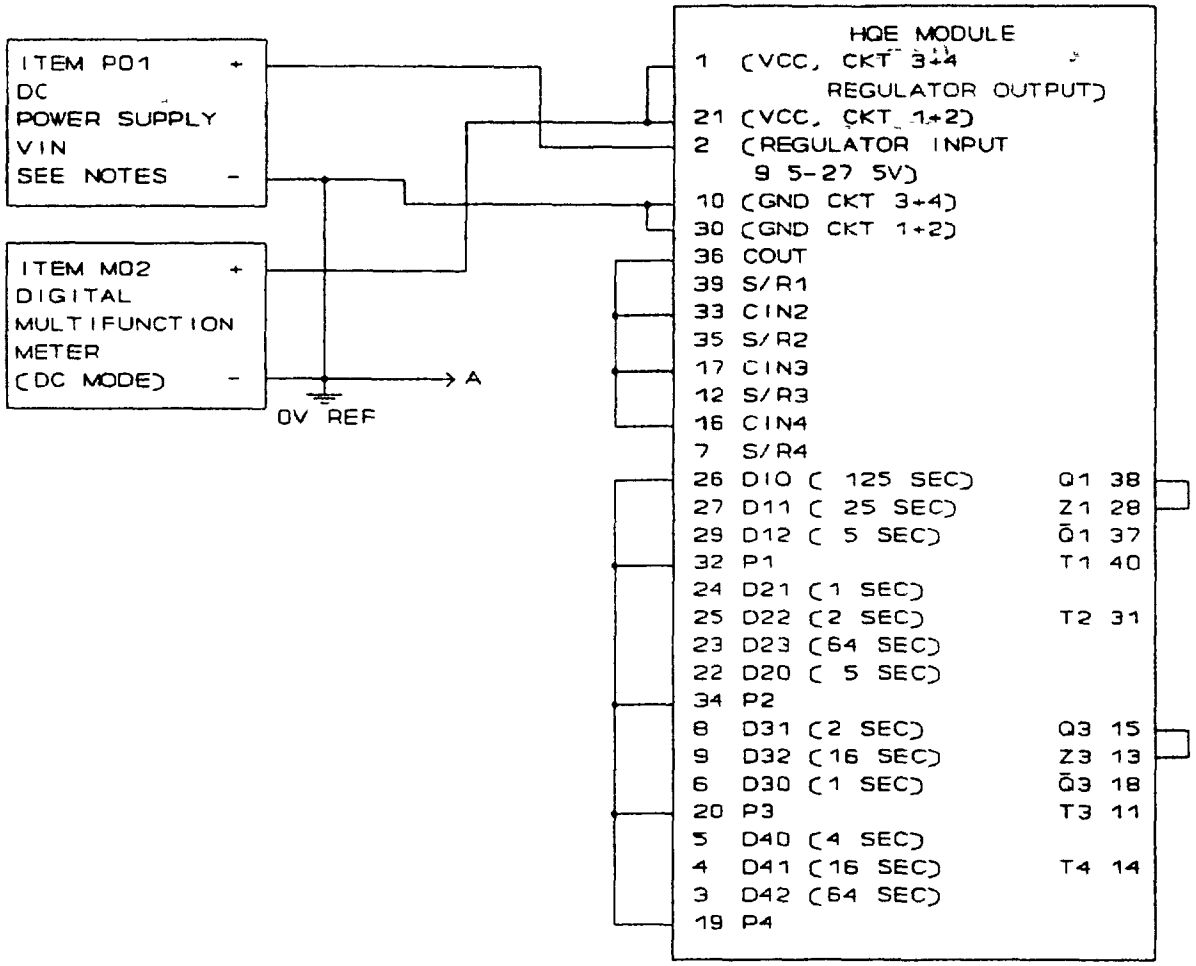
V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

CAUTION:
Do not isolate chassis of ammeter X01 from earth ground.

FIGURE 19. 2 input current, low level, test circuit for table XV.

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CAUTION:
Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 23. Power supply, low level and high level, regulation test circuit for table XIX.

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WARNING: (Continued)

Make power supply voltage adjustments before energizing test circuit.

NOTES:

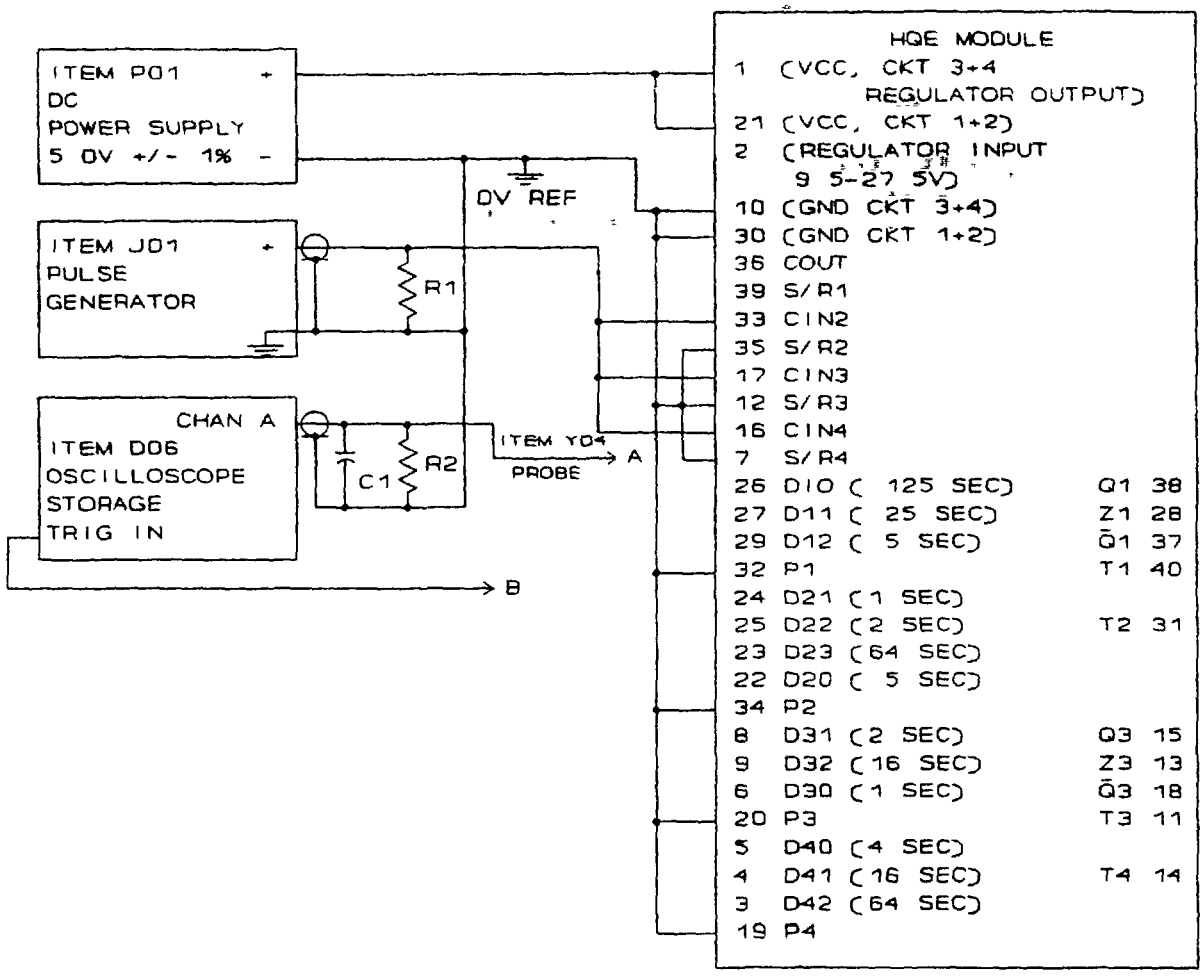
1. For test condition Q1, V_{IN} = 9.5 V, +1%, -0.

2. For test condition Q2, V_{IN} = 27.5 V, +0, -1%.

3. For test condition Q2, measure output 0.13 seconds after pin A is connected.

FIGURE 23. Power supply, low level and high level, regulation test circuit for table XIX - Continued.

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CAUTION:
Failure to comply with the following procedures will result in irreversible module damage.

All test equipment must be grounded.

The module shall not be removed from or inserted into a hot socket.

V_{CC} must be applied before applying input signals or conditioning voltages.

All input signals and conditioning voltages must be removed before V_{CC} is disconnected.

FIGURE 31. Timer 2-4 output transition test circuit for table XXIV.

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NOTES: (Continued)

R1 = 51 Ω ± 5%, 1/4 W.
R2 = 200 kΩ ± 1%, 1/8 W.
C1 = 50 ± 5 pF (including probe and fixture capacitance).

FIGURE 31. Timer 2-4 output transition test circuit for table XVIV - Continued.

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